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Software Requirements Specification

For



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Revision History

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1. Introduction

1.1 Purpose

Purpose of the Sanad System: The vision of the "Sanad platform" is to allow individuals with disabilities to have full and independent access to their daily activities, including recreational, educational, employment, and social activities, through integrated digital tools, including: an interactive map of accessible sites, a forum for empowerment support, appointment and escort services, e-learning and accessible job opportunities, and a library of assistive products. All of this is intended to promote themselves in effectivity into their lifestyle and full participation.

This goal is supported using research evidence that documents the impact use of digital technologies and assistive technologies has on enhancing quality of life and social participation for individuals with disabilities. For example, "assistive technology (AT) provides compensatory or supportive tools for individuals with disabilities, thereby increasing independence and an ability to participate in everyday life and community activities."

Additional research suggests that "digital technologies can serve as an important means of facilitating the social integration of people with disabilities into society and particularly for individuals who struggle to participate in community life." Accordingly the "Sanad platform" seeks to realize this vision by offering a suite of services that are appropriate for people with disabilities, linking the technological dimension (map, services platform and library of technologies) with the social dimension (forum and stories of inspiration) and the professional/educational dimension, in order to achieve three main outcomes:

1. Promoting access to physical locations, service spaces, and participation.
2. Improving communication and information-sharing between people with disabilities and the community.
3. Building and improving access to learning and employment options that lead to independence, economic participation, and community inclusion.

In this way, the platform is aiming to serve as a "unified digital environment" for people with disabilities to achieve participation and equality; this is conceived to be achieved in the technical sense and from a social and economic perspective.

1.2 Scope

The Sanad Platform offers an online accessibility system that has the potential to empower individuals with disabilities by providing them with an easy and independent means of accessing services, places, and opportunities. The platform offers an online community for individuals with disabilities that provides them with vital resources such as an accessible online interactive map of accessible places, appointment scheduling with helpers and interpreters, an online community arena, an online library of assistive devices, as well as training and employment opportunities. The Sanad Platform has the capacity to improve the lives of individuals with disabilities by providing them with the opportunity to enjoy full independence and participation. Users of the Sanad Platform can enjoy online accessibility via web as well as mobile.

Through collaboration, the project team can create an integrated platform that incorporates accessibility functionality, socialization, as well as professional growth. The proposed system entails functionalities such as registering users, authenticated login functionality, mapping accessible services within the users' geographic location, a community forum for discussions and exchange of views on personal experiences of accessibility services, services for volunteering and

interpreter services, besides an educational and jobs hub enabling users to connect with training sessions and jobs. The system also offers a digital library aspect that exhibits assistive devices and technologies such as screen readers, smart canes, and speech command devices. In contrast to the community or service platform that may serve the needs of the general public, Sanad deals exclusively with digital accessibility for the disabled community.

The Sanad Platform provides multiple advantages such as increased levels of independence, inclusion, and communication for users, families, as well as service providers. The platform encourages collaboration and increases awareness of accessibility. In addition to these benefits it should ensure that it provides security, accessibility and maintainability by implementing encrypted data protection by design.

The overall objective of the Sanad Platform is to offer individuals with disabilities an accessible and easy to use space that connects them with services and opportunities that increase levels of self reliance. helping them achieve greater self reliance and participation in their communities.

1.3 Definitions, Acronyms, and Abbreviations

Terminologies are listed in the table below:

Terminology	Definitions
Sanad	Sanad Platform is a unified digital system that provides accessible digital services to empower people with impairments.
Musanada	An alternative term for the same accessibility idea as Sanad is an Arabic word that means "support."
Version	A system or document version number refers to a certain edition or release.
Disability	A physical, mental, cognitive, or developmental condition that limits a person's ability to perform certain activities or intricate with the world.
Digital tools	Applications, software, websites, and other computer or electronic resources are used to perform task, enhance functions, and facilitate processes.
Recreational	Something done for pleasure, fun, or enjoyment in one's spare time, rather than for work or professional purposes.
ER Diagram	Visual representation of data entities and the relationships between them
Sequence Diagram	A diagram representing exchange interactions over time and with accuracy
Video Communication APIs	Application program interface for facilitating video calls.
Employment or Training Database	External database for jobs\education.
Sign Language Interpretation	Translating spoken language to sign language.
Volunteer Companion	Volunteer providing assistance.
Interactive Map	Interactive digital map for user interface.

Accessibility Layer	Overlay feature highlights accessible elements.
GPS Tracking	Monitoring location in real time.
End-to-End Encryption	Secure transmission of data.
Input Validation	Checking for user input.
Data Sanitization	Cleaning data to avoid or prevent harm.
Containerization	Packaging software into containers.
Cloud Migration	Moving to a cloud-based platform.
Failover	Switching to automatic backup.
Disaster Recovery Plan	System recovery strategy.
Scope Creep	Changes in a project without control.
Change Control Board	Groups which are approving of the changes.

Acronyms	Meanings
SRS	Software Requirements Specification a document outlining the software's objectives, features, and limitations.
AT	Assistive Technology – tools that aid persons with disabilities
API	An Application Programming Interface facilitates interaction between different software applications, allowing them to communicate effectively.
IoT	The Internet of Things refers to a network of interconnected devices that share data.
IEEE	Institute of Electrical and Electronics Engineers – a professional organization dedicated to engineering standards.
ISO	The International Organization for Standardization is responsible for the development and publication of international standards.
IEC	The International Electrotechnical Commission establishes standards for electrical and electronic technologies.
CRPD	Convention on the Rights of Persons with Disabilities – a United Nations treaty that advocates for the inclusion of individuals with disabilities.
MPT	Matching Person and Technology a framework for assessing the suitability of assistive technologies.
WHO	World Health Organization UN agency for global public health.
UN	United Nations international organization promoting peace and cooperation.
ID	Identifier (Article ID) unique code for academic or technical documents.
Vol.	Volume finds the volume number in a journal.
No.	Number shows the issue number of a publication.
PP.	Pages writes down page range in references.
CIS	Computer Information Systems field studying computing and information

	systems.
CS	Computer Science study of algorithms, programming, and systems.
Dr.	Doctor academic or professional title for individuals with a doctoral degree.
Std	Standard officially recognized technical specification
UI	User Interface visual part of a system users interacts with.
OTP	One-Time password. Which is a temporary verification code sent to the user via SMS or email to confirm identity during login, registration, or password reset. The code is valid for only one session and expires after a short period for security purposes.
2FA	Two-Factor authentication. Which is a security method that requires the user to verify identity using two steps, usually a password and OTP code sent to phone number or email.
HTTPS	Hypertext transfer hypertext transfer. Which is a secure communication protocol used to protect data transferred between the user and the server by encryption.
XSS	Cross-Site scripting. Which is a security vulnerability where an attacker injects malicious scripts into web pages to steal data or hijack user accounts.
UUID	Universally unique identifier. Which is a 128-bit unique value used to uniquely identify records in the system (user ID) without conflicts.
Enum	Enumeration type. Which is a data type consisting of a set of predefined values. Used to avoid invalid data entry.
SMS	Short message service. Which is a text messaging service used for sending OTP codes or notification messages to users' mobile phones.
CVS	comma separated values.
JSON	Java script object notation.
APIs	Application programming interface.
GPS	Global positioning system.
DRP	Disaster recovery plan.
CCB	Change control board.
WCAG	Web content accessibility guidelines.
PDPL	Personal data protection law.
HIPAA	Health insurance portability and accountability act.
HTML	Hypertext markup language.
CSS	Cascading style sheets.
JavaScript	Programming language for web interactivity.
React	JavaScript library for UIs.
JWT	Json web token.
IOS	Apple mobile OS.
Android	Google's mobiles OS.
W3C	Organization for web standards
DGA	Saudi authority for digital services.
MHRSD	Saudi government ministry.
AWS	Cloud platform.
PayPal	Payment platform.
Stripe	Payment processing

QA	Ensuring software quality.
CI\CD	Continouns intergration\ deployment.
ITIL	IT service mangement framework.
COBIT	IT goverance framework.

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1.5 OVERVIEW

This part presents a short summary of the Software Requirements Specification (SRS) document for the project called "Sanad" platform. This document defines all functional and non-functional requirements of the system and acts as a formal agreement between the client and development team. The document is structured in those parts:

- Section 2 (Overall Description): Gives a full description of the product, the capabilities of the intended users, constraints, and broader overall assumptions about the project.
- Section 3 (Specific Requirements): It is the main section where all requirements are described in detail, including external interfaces, specific functional requirements (the map, services and forum), details about data categories, and non-functional requirements (performance, security and reliability)
- Section 4 (Analysis Models): Includes the analysis models of the project, such as an Entity-Relationship Diagram (ER Diagram) and sequence diagrams that represent logical representations of the system.
- Section 5 (Change Management Process): Identifies formal processes for both managing and controlling any future changes made to the requirements of this system.

2. Overall description

2.1 Product perspective

The "Sanad" platform is an all-inclusive web-based and mobile-friendly system created to help people with disabilities by making their everyday lives easier and increasing their social inclusion.

The platform is a stand-alone system that communicates with a number of external services, including :online mapping APIs (for the visualization of accessible locations) video communication APIs (for remote sign language interpretation) and employment or training databases (for the provision of appropriate employment and educational opportunities).

Each module of the system, such as the Assistive Technologies Library, Community & Support, Services & Appointments, Learning & Employment and Accessible Places Map, functions in a shared environment and communicates with one another via a single backend.

A common user interface and authentication mechanism are shared by all modules, guaranteeing uniformity, usability and a seamless user experience throughout the platform.

2.2 Product functions

An integrated electronic platform is being designed for people with disabilities to facilitate their daily lives and enhance their integration into society. The platform's most important services include:

1- Accessible Places Map: An interactive map showing accessible places such as cafes, restaurants, universities, and hospitals. This map includes filters to identify accessible features (such as wheelchairs and accessible restrooms). Platform users can add new places and share their experiences.

2- Community and Support: The platform has a forum where users can talk about issues and share tips and stories. Posts are reviewed to make sure the forum stays safe.

3- Services and Appointments: Through the platform users can book appointments with remote sign language interpreters. They can also use the "Volunteer Companion" service to request a companion for a trip or appointment. The system includes: a scheduling tool that facilitates organizing and following up on appointments.

4- Learning and Work: The platform offers online training courses suitable for people with disabilities and links to employment portals to provide accessible jobs.

5- Assistive Technology Library: Users can access a range of assistive tools including: speech devices, screen readers, and smart canes. They can also request assistance in obtaining devices.

6- User Management and Access Features: The platform provides secure login with different access levels (administrator, volunteer and user). Alerts and reminders about platform updates and new appointments are also provided.

2.3 USER CHARACTERISTICS

This section outlines the primary roles (actors) who will interact with the "Sanad" system, and describes the critical qualities and needs for each role with an emphasis on the importance of accessibility.

Actor	Description	Accessibility
Client User	The users are people with impairments who are accessing information, services, or work facilitation. They require:	The application must be compliant with WCAG 2.1.
	• A simple and easy to utilize interface • A way to browse with assistive tools screen readers and alternative keyboards. • A high level of accuracy in identifying geographic locations.	The application must have high contrast mode and changing font size features.
Service Provider	This involves volunteers and interpreters who deliver escort or interpretation services. They require:	The application of acceptance/rejection should have responsive decision-making with realistic input fields.
	• A strenuous and documented registration and worker verification process. • An expedient notification system to receive bookings and matching determinations in real-time. • A real-time GPS tracking system while escorting.	
Admin	Tasked with overseeing the platform and content, resources, and users. They require:	The administrator interface will be usable from standard web browsers and will support keyboard navigation.
	• An efficient and secure administrative dashboard. • Effective tools, with edits and storage options to review content (reviews, new listings, and forum posts). • The ability to create reports on performance and security.	

2.4 GENERAL CONSTRAINTS

These constraints describe the obligatory aspects the design and development team must comply with, which are not tangibly tied to functionality, but include the operating environment and quality:

Type	Detailed restrictions
Organizational and legal	GC- L1 Complying with the Saudi Arabia Personal Data Protection Law (KSA PDPL) in full to protect sensitive data from users (location, health status, records of service).
	GC- L2 The videoconference system must have confidentiality and privacy requirements that meet standards for handling health/personal data (like references to HIPAA standards or local equivalent).
Technical standards and limitations	GC-T1 All interfaces (both web and mobile) must comply with Web Content Accessibility Guidelines (WCAG) standards as a minimum requirement.
	GC-T2 Both English and Arabic languages must be available.
Mandatory technologies	GC-T3 HTTPS protocol must be used for all communication between the client and the server, and encryption must be used for all stored data.
	GC-T4 Mobile application development should be done using a cross-platform framework (e.g., Flutter or React Native). This is to ensure that a single codebase can compile the app for both iOS and Android.

2.5 ASSUMPTIONS AND DEPENDENCIES

This section outlines the external factors that are out of the control of the project team but are critical for the success of "Sanad," along with the assumptions of the requirements.

2.5.1 Assumptions

Assumption	Impact on the project
AS-1 Accessing data: We expect that the requisite APIs to extract job and training course data for people with disabilities, from outside platforms, will be available either free or at a nominal cost.	In the event that these APIs are unavailable or excessively costly, it may be required to input the information manually which affects effort and scalability.
AS-2 User intent: We will take for granted that registered users (particularly volunteers and translators) have a legitimate intent to provide the service and that their basic information is correct in the appropriate space.	If this assumption fails, it may necessitate increased background checks and training to develop the administrator system (Admin).
AS-3 Network Coverage: It is assumed that you have a valid internet coverage (4G/Wi-Fi) with enough quality for the video calling software to be able to interpret sign language.	If the connection is weak, the NFR-R3 requirement (visual communication quality) may fail.

2.5.2 Dependencies

Dependencies	Impact on the project
DP-1 Mapping Service: The platform is wholly dependent on an external map service provider (Google Maps API, or similar) for location services.	Modifications in the pricing or terms of service associated with map APIs will impact operating costs for the platform.
DP-2 Payment Gateway: Whenever paid services are required (such as purchasing), the system relies on successful integration with a locally approved external payment gateway(. (Visa Payment or mada)).	Payment transactions depend on reliability of the external gateway and compliance with PCI DSS requirements.
DP-3 Verification of the content: The system counts on official or professional organizations to verify the accuracy of what is provided as configuration information for some sensitive sites (such as hospitals).	Delays in that verification would result in delays in the sites being deployed in the live database.

3. Specific requirements

3.1 External interface requirements

The external interface requirements define how the “Sanad” platform interacts with external entities, including users, devices, and other software systems. Given that the platform aims to support and empower individuals with disabilities in their daily lives, these requirements focus on providing accessible and effective means for users to utilize various platform services, such as the Accessible Places Map, appointment and escort services, community support, learning and employment opportunities, and the Assistive Technology Library.

These requirements encompass all interaction points for data input and retrieval, ensuring secure and reliable communication with external services such as mapping APIs, video communication systems supporting sign language, and employment or training platforms. Additionally, the external interface requirements consider support for assistive devices and technologies that enable users with disabilities to interact flexibly with the system.

Overall, the external interface requirements ensure continuous and seamless connectivity between the platform, its users, and supporting systems, enhancing the platform’s goal of enabling independent participation and access to services for people with disabilities.

3.1.1 User interfaces

- **Home Page Interface:**

When a user opens the Sanad Platform, the home page provides access to all main services through a simple and inclusive layout.

- **Navigation Options:** Links to key sections such as Interactive Map, Community Forum, Services, Education & Jobs, and Assistive Technologies.
- **Accessibility Features:** High-contrast mode, adjustable text size, and text-to-speech options for visually impaired users.
- **Search Bar:** Allows users to find locations, services, or topics quickly by keywords or categories.
- **Highlights Section:** Displays featured services, recent discussions, and success stories to encourage engagement.

- **Login and Registration Interface:**

- **Login:** Registered users can securely access their accounts using email and password.
- **Sign Up:** New users can register by providing personal information, disability type (optional), and communication preferences.
- **Social Login:** Login available via Google or Apple ID for convenience.
- **Password Recovery:** Secure reset option through verified email.
- **Error Messages:** Clear messages for incorrect credentials or unverified accounts.

- **Interactive Map Interface:**

- **Map Display:** Shows accessible public facilities such as hospitals, schools, and restaurants with accessibility indicators (e.g., wheelchair access, Braille signage).
- **Filters:** Filter by category, accessibility type, or location.
- **Details Panel:** Provides details about each location, including contact info, ratings, and accessibility features.
- **Navigation Support:** Integration with Google Maps for route guidance.

1. Community Forum Interface:

- **User Posts:** Users can create posts, share experiences, and reply to others.
- **Categories:** Divided into topics such as “Daily Life,” “Education,” “Employment,” and “Accessibility Issues.”
- **Moderation Tools:** Report or block inappropriate content to maintain a positive environment.
- **Likes and Comments:** Simple engagement tools to encourage participation.

2. Volunteer and Interpreter Services Interface:

- **Request Form:** Allows users to request a volunteer or interpreter, specifying the type of assistance, time, and location.
- **Service Matching:** The system suggests available volunteers/interpreters based on user needs.
- **Status Tracking:** Shows whether a request is pending, accepted, or completed.
- **Notifications:** Users receive real-time updates through alerts or email.

3. Education and Job Opportunities Interface:

- **Search and Filter:** Users can browse training courses or job openings suitable for different disability types.
- **Application Portal:** Enables users to apply directly or contact the organization offering the opportunity.
- **Recommendation System:** Suggests relevant opportunities based on user profile and interests.
- **Feedback Section:** Allows users to rate courses or job programs.

4. Assistive Technology Database Interface:

- **Categories:** Displays technologies such as screen readers, speech-recognition software, or smart canes.
- **Product Details:** Includes descriptions, images, and vendor links.
- **User Reviews:** Community feedback on usability and performance.
- **Comparison Tool:** Users can compare multiple tools side by side.

5. User Profile Interface:

- **Profile Management:** Users can update personal information, communication preferences, and disability-related needs.
- **Activity History:** Displays a list of previous requests, posts, and saved resources.
- **Accessibility Settings:** Customizable features like preferred language, display mode, and notification preferences.
- **Account Deletion:** Allows users to deactivate or permanently delete their account.

6. Admin Dashboard Interface:

- **User and Content Management:** Admins can monitor, approve, or remove posts and accounts.
- **Service Oversight:** View and manage volunteer and interpreter requests.
- **Reports and Analytics:** Displays statistics such as platform activity, popular services, and system performance.
- **System Alerts:** Notifications for errors, maintenance needs, or abnormal activities.

3.1.2 Hardware Interfaces

7. **Device:** Windows, macOS, or Linux workstation.
8. **Processor:** Minimum Intel i3 (1.8 GHz or higher).
9. **RAM:** 4 GB or more.
10. **Storage:** 10 GB of free space.
11. **Peripherals:** Standard keyboard, monitor, and pointing device.

For End Users:

- **Device:** Desktop, laptop, tablet, or smartphone with Internet access.
- **Accessibility Devices Supported:** Screen readers, speech recognition tools, Braille displays, adaptive keyboards, and voice-controlled devices.

3.1.3 Software Interfaces

8. **Frontend:** HTML, CSS, JavaScript (React framework).
9. **Backend:** Node.js with Express.js framework.
10. **Database:** MySQL for structured data storage.
11. **APIs:** RESTful APIs for communication between modules.
12. **Integrations:**
 1. Google Maps API for location services.
 2. PayPal or Stripe for secure service donations (optional).
 3. Cloud services (AWS or Google Cloud) for data and image hosting.
13. **Security:** HTTPS protocol, JWT authentication, and encryption for user data.
14. **Accessibility Standards:** WCAG 2.1 compliance to ensure usability for users with disabilities.

3.1.4 Communications Interfaces

- **Network Protocol:** HTTPS for secure communication between client and server.
- **Data Transmission:** JSON format through RESTful API.
- **Real-Time Updates:** WebSocket integration for live notifications and chat features.
- **Third-Party Services:** Secure communication with Google Maps, cloud storage, and email services.
- **Browser Compatibility:** Chrome, Edge, Firefox, and Safari (latest versions).
- **Error Handling:** System alerts for connection loss, server errors, or invalid requests.

3.1.5 Operating System and Versions

1. Mobile Application

- **iOS 12 and above**
- **Android 10 (API level 29) and above**

2. Web Application

- **Windows 10 / 11**
- **macOS Catalina (10.15) and above**

3.2 FUNCTIONAL REQUIREMENTS

This section describes the specific procedures and processes that the Sanad platform must perform to meet user needs. The requirements have been grouped according to the main function.

3.2.1 <User Registration and Profile #1>

This section describes the functional requirements for users (individuals with disabilities, their families, volunteers, interpreters and service providers) account creation, authentication, and profile management for the Sanad Platform. These requirements ensure that users can register, sign-in, and maintain their profile information securely and efficiently.

Requirement	Priority
FR1.1 The system should allow the user to create an account (as a client, volunteer or interpreter) using an email or phone number and password.	High
FR1.2 The system should allow the user to optionally specify the type of disability (visual, motor or auditory) to enable interface customization.	High
FR1.3 The system should allow the user to log-in using their registered data.	High

FR1.4 The system should allow the user to update their profile, manage their data and change their password.	Medium
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3.2.2 <Map Search and Rating Management #2>

This section describes the functional requirements for the interactive accessibility map. The system will allow users to search for places of public accommodation, filter them by accessibility features, and view detailed information about each specific location. Users will also be able to add new places to the map and report any inaccuracies. Moderators will be able to review those additions, as well as the data shared above, for accuracy and reliability.

Requirement	Priority
FR-2.1 Search and Filters: Users shall search for places by entering keyword search terms and select filters by accessibility or by category and the results shall show both in the map view and list view.	High
FR-2.2 Place Details: When the user selects a marker, the system shall present a details card that includes accessibility badges, photos, average rating and most recent reviews.	High
FR-2.3 Ratings and Reviews: Authenticated users shall always submit ratings/reviews with optional photos and the average ratings shall update dynamically.	High
FR-2.4 Moderation: New places and reviews will be marked Pending review. Approve or reject and/or request further information, moderators will have the ability to do this. All moderator actions will log an audit trail.	High
FR-2.5 Reporting: Users shall report content if it is inaccurate or inappropriate. The user will be able to view the status of the report until resolved and will be notified a decision has been made.	Medium

3.2.3 <Community and Content Management #3>

This part explains the functional requirements for 1- engaging with the community .2- participating in forums. 3- creating content. 4- engaging with users on the platform. These requirements enable users to create posts, reply to posts, report inappropriate content and engage in a manner that is safe within the community environment.

Requirement	Priority
FR-3.1 Forum Provision: The system will have an organized electronic forum, the Categorized Forum, where registered users can add new topics and reply to existing topics.	High
FR-3.2 Content Reporting: It shall allow users to report any inappropriate, offensive, or irrelevant content that has been posted within the forum.	High
FR-3.3 Inspirational Stories Section: It should also have a section for "Inspirational Stories" where only stories that have been reviewed and accepted by the moderators can be published.	Medium
FR-3.4 User Notifications: The system will notify the users whenever a reply to a topic created or followed by them is posted.	Medium

3.2.4 <Service and Appointment Booking #4>

The system should allow users to book various services such as sign language interpreters or volunteer companions, with the ability to schedule appointments with appropriate details for each type of service.

Requirement	Priority
FR-4.1 Booking sign language interpreters: The system shall allow the user to view and book an interpreter appointment (date and time).	High
FR-4.2 Video call (Encrypted): The appointment between the user and interpreter (video call) shall be held in an encrypted, private video room.	High
FR-4.3 Request volunteering escort: The system shall allow the user to enter details about the request (route, time, purpose).	High
FR-4.4 Matching volunteers: The system shall match users' request to available volunteers based on geographic location and notify them of the request in order to accept and assist.	Medium

FR-4.5 Reschedule: The system shall allow the user the ability to reschedule an appointment prior to the time of service, if availability permits.	Medium
FR-4.6 Cancellation: The system shall allow the user the ability to cancel a booked appointment (within a specified time frame before the appointment---up to 24 hours before).	Medium
FR-4.7 Appointment Status: The system shall allow the user to view what the current status is (Pending, Accepted, Rejected, Completed).	Medium

3.2.5 <Learning, Employment, and Technology #5>

This section sets out the functional requirements for the access of opportunities for learning, employment, and assistive technology resources on the platform. Users will be able to navigate through course offerings and job postings, view more detailed information, save items for future reference, and open an external link to register or apply. These functions are intended to help users improve their skills, find jobs that are right for them, and discover technology that assists them in their everyday lives.

Requirement	Priority
FR-5.1 View Learning Opportunities: The system will provide the user the ability to search available workshops, training courses, and learning programs.	High
FR-5.2 Course / Program Details: The system will provide full details about courses (provider, duration, cost, and registration link).	High
FR-5.3 Job Listings Browsing: The system will allow the user to search job opportunities and filter them by category, disability support, and/or location.	High
FR-5.4 Job / Opportunity Details: The system will provide details about the job opportunity (requirements, location, and apply link).	Medium
FR-5.5 Assistive Technology Library: The system will allow browsing assistive technology items (devices, apps, and tools) that support accessibility needs.	Medium
FR-5.6 Save Items / Favorites: The system will allow the user to save courses, jobs, or assistive technology items into a personal "Saved Items" list.	Medium
FR-5.7 Open External Link: The system will allow users to open an external link to register for a course, or apply for a job.	Medium

3.2.6 <Admin and Moderation Management #6>

This part explains the functional requirements for 1-the administrators and moderators that must review content 2-manage the reporting process 3-monitor user activity and improve safety and quality of the overall experience on the platform. These requirements allow the moderation team to manage posts, approve submissions and monitor system health.

Requirement	Priority
FR-6.1 Admin Dashboard: The administrative dashboard of the system should provide features for moderators to review newly added places and classifications that are decided upon their approval or rejection and later publish them on the map.	High
FR-6.2 Content Monitoring & Actions: The system shall permit moderators to observe the forum and take appropriate actions against injurious content or violating accounts, such as warnings, temporary suspension, and deletion.	High
FR-6.3 Performance & Analytics Reports: It shall allow moderators to generate various reports on performance and analytics: number of place bookings, service popularity, forum activity.	Medium
FR-6.4 Moderators & Volunteers Account Management: The system should provide the moderator with the ability to manage both volunteers and other moderators' accounts, such as reviewing their contributions and performance.	High

3.3 USE CASES DIAGRAM

The use case diagram provides a high-level functional overview of the Sanad Platform. It defines the User as the system's primary actor and demonstrates how the user interacts with core system functions. Including registration, log-in, profile management and password recovery. This diagram supports understanding of the system's functional scope and user-centered workflow.

3.3.1 <User Registration and Login>

The system allows the user to create account by entering their email or phone number and password with an optional disability type to make the platform easier to use by people with disabilities, or log-in by verifying their email or phone number and password as shown in figure [3.3.1].

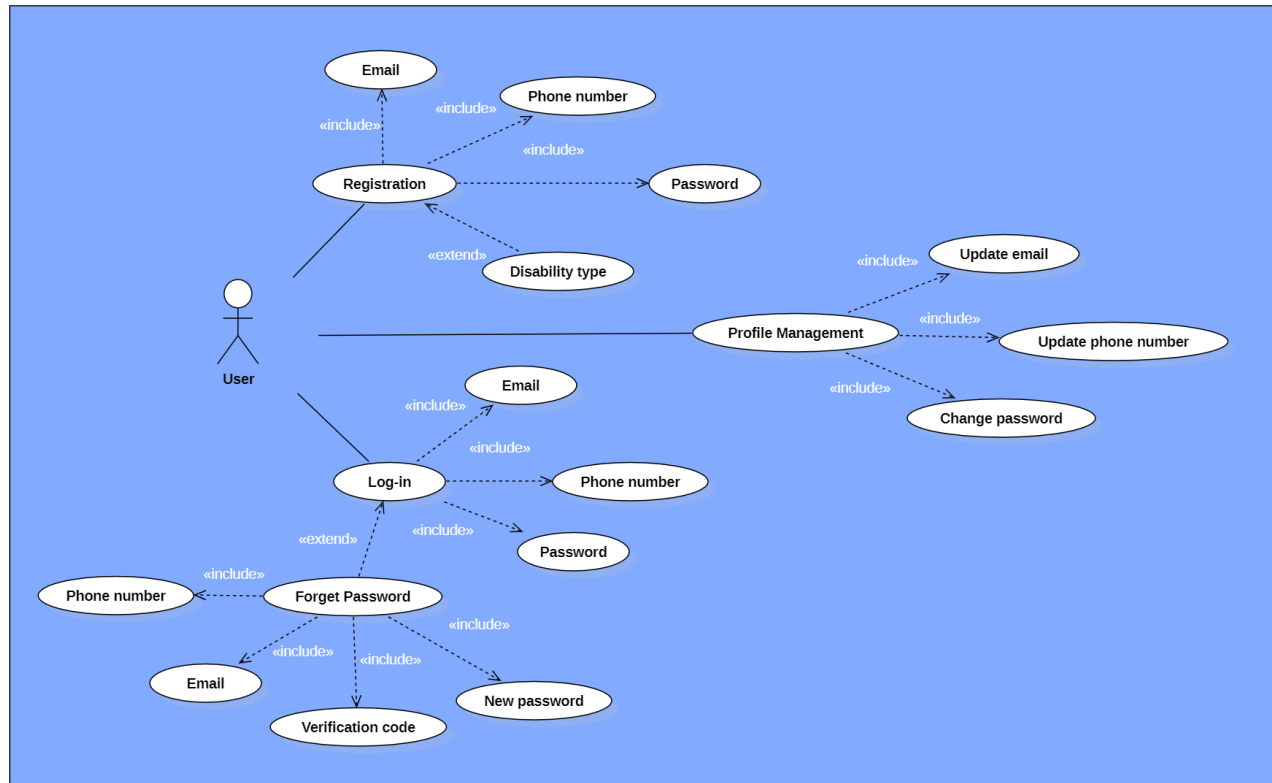


Figure 3.3.1 – User Registration and Log-in

Sanad Platform: User registration	
Actors	Individuals with disabilities, their families, volunteers and translators.
Description	The user creates a new account within the Sanad platform by providing personal information such as an email or phone number and creating a password. The user may also optionally specify their disability type to allow the system to customize the interface for accessibility needs. The registration process also requires the user to select their account type (client, volunteer, or translator), after which the system validates and stores the provided data. Once completed, the system confirms the account creation and grants the user access to log in and begins using the platform.
Data	2 Email or phone number. 3 Password. 4 Account type (Individuals with disabilities, their families, volunteers and interpreters). 5 Disability type (optional).
Stimulus	The user selects "Create Account" from the application.
Response	The system creates the user account and confirms successful registration.
Comments	The disability type is optional and used for accessibility interface adaptation.

Sanad Platform: User log-in	
Actors	Individuals with disabilities, their families, volunteers and translators.
Description	The user accesses the platform by entering their previously registered email or phone number along with their password. The system verifies the login credentials, and if the information matches stored data, the user is granted access to their personal profile and the services offered by the platform. If the user forgets their password, they may choose the “Forgot Password” option to recover or reset their login details before proceeding.
Data	- Email or phone number. - Password.
Stimulus	The user selects “Log-in” and enters their login credentials.
Response	The system verifies the credentials and grants access to the profile and platform services.
Comments	“Forget Password” is an extension scenario that occurs if the user cannot remember the password.

Sanad Platform: Profile management	
Actors	Individuals with disabilities, their families, volunteers and translators.
Description	After logging into the platform, the user may view and update their personal profile information. This may include modifying their contact details, changing the disability classification, or updating personal preferences related to accessibility. The user may also manage settings related to visibility and communication. Once the changes are saved, the platform updates the stored records, ensuring that future system interactions reflect the modified profile.
Data	6 Email or phone number. 7 Password. 8 Disability type.
Stimulus	The user selects “Profile” then “Manage Profile” after logging in and update their information.
Response	The system updates the user profile and saves the new information.
Comments	The user must be authenticated before accessing this use case.

Sanad Platform: Change password	
Actors	Individuals with disabilities, their families, volunteers and translators.
Description	Within the profile management section, the user may choose to change their password for security or personal preference reasons. To complete this process, the user must enter their current password before specifying a new one. The system validates the inputs and updates the stored password if the authentication is correct. The user then receives confirmation that the new password has been successfully applied.
Data	• Email or phone number. • Password.
Stimulus	The user selects “Change Password” if they forget their password.
Response	The system allows them to change their password then validates and updates the password successfully.
Comments	Users must be logged in to access this feature.

3.3.2 <Map & Location Management>



Figure 3.3.2 Map & Location Management

Sanad Platform:Map Browsing & Layers	
Actors	User
Goal	Allow users to explore the map and enable or disable accessibility layers for better visibility of accessible locations.
Preconditions	• Map SDK must be active. • Optional location permission. • Accessibility data should be available.
Trigger	User opens the map tab from the interface.
Main Flow	1) System loads and centers the map. 2) User pans or zooms. 3) User toggles accessibility layers. 4) System displays updated markers. 5) User selects a specific location for more details.
Alternative Flows / Exceptions	A1) GPS disabled → show default city. A2) No data → display 'No data available' message.
Postconditions	Map state (zoom and layer preferences) is saved for the next session.
Business Rules	Show verified and reliable locations first.
Non-Functional Requirements (NFR)	Map should load within ≤ 2 seconds.
Notes	All location tracking follows the privacy and data retention policy.

Sanad Platform: Search & Filters	
Actors	User
Goal	Enable users to search accessible places using keywords and filters such as type, rating, or distance.
Preconditions	• Search index must be active and synchronized.
Trigger	User activates the search bar from the map screen.

Main Flow	1) User types in keywords. 2) User applies filters. 3) System returns filtered results.
Alternative Flows / Exceptions	A1) If no results found → suggest expanding search radius.
Postconditions	Applied filters are saved for the next session.
Business Rules	Ranking priority: verified > rating > distance.
Non-Functional Requirements (NFR)	Response time should be $\leq 800\text{ms}$.
Notes	Search operations comply with privacy guidelines and minimal data retention.

Sanad Platform: Submit Place / Accessibility Review	
Actors	User
Goal	Allow users to add new accessible locations or submit accessibility reviews for existing ones.
Preconditions	• User must be authenticated. • Anti-abuse system enabled.
Trigger	User selects 'Add Place' option from the map menu.
Main Flow	1) User fills out submission form. 2) System validates the data. 3) Entry saved as 'Pending Review'.
Alternative Flows / Exceptions	A1) If duplicate detected → system suggests merging entries.
Postconditions	Submission stored and queued for admin moderation.
Business Rules	Accuracy threshold: location precision ≤ 30 meters.
Non-Functional Requirements	Offline draft support for users without internet.

(NFR)	
Notes	All submissions respect privacy and validation policies.

Sanad Platform:Report Issue / Request Edit	
Actors	User / Moderator
Goal	Allow users to report incorrect or outdated accessibility information and suggest corrections.
Preconditions	• User must be logged in. • Moderation tools must be active.
Trigger	User clicks 'Report' or 'Suggest Edit'.
Main Flow	1) User selects issue type. 2) Adds explanation or evidence. 3) System logs a support ticket.
Alternative Flows / Exceptions	A1) Repeated reports automatically increase issue priority.
Postconditions	Ticket saved for review and resolution by moderators.
Business Rules	Moderator approval required before publishing edits.
Non-Functional Requirements (NFR)	Response time for reported issues ≤ 24 hours.
Notes	All reports comply with privacy policy and moderation standards.

Sanad Platform: Accessible Route Planning	
Actors	User
Goal	Provide wheelchair-friendly routing and navigation between two points.
Preconditions	• GPS must be on. • Routing engine must support accessibility constraints.

Trigger	User selects 'Directions' from a chosen place.
Main Flow	1) User chooses transportation mode. 2) System suggests accessible routes. 3) User starts navigation.
Alternative Flows / Exceptions	A1) If no accessible route found → offer alternative routes.
Postconditions	Chosen route stored for quick access in future sessions.
Business Rules	Avoid stairs and steep paths by default.
Non-Functional Requirements (NFR)	System reroutes within ≤ 2 seconds after deviation.
Notes	Routing follows user consent and privacy requirements.

Sanad Platform: Live Location Sharing	
Actors	User, Volunteer/Interpreter
Goal	Enable users to securely share live location for easier coordination with volunteers or interpreters.
Preconditions	• Mutual consent required. • Active internet connection.
Trigger	User taps 'Share Location'.
Main Flow	1) User selects contact to share with. 2) System generates secure token. 3) Recipient can view live location on map.
Alternative Flows / Exceptions	A1) If recipient has no app → open secure web link.
Postconditions	Session expires automatically after a defined duration.
Business Rules	Shared location expires after 30–60 minutes.

Non-Functional Requirements (NFR)	Encrypted; updates every 10–20 seconds.
Notes	Live location sharing strictly follows data protection and consent rules.

3.3.3 <Appointment and Volunteer Management>



Figure 3.3.3 -Appointment and Volunteer Management

Sanad Platform: Interpreter Appointment Booking	
Actors	User, Interpreter
Goal	Enable users to find, book, and confirm interpreter appointments.
Preconditions	The user must be logged in, and interpreters should have available time slots.
Trigger	User chooses the "Interpreter Booking" option.
Main Flow	1) Enter appointment details. 2) System displays available interpreters. 3) User selects preferred interpreter. 4) User confirms booking. 5) Notifications are sent to both parties.
Alternative Flows	A1: No interpreters available. A2: Scheduling conflict occurs. A3: Video session fails to start.
Postconditions	Appointment is successfully created, and notifications are delivered.
Business Rules	Bookings must follow platform time and availability policies.

Sanad Platform: Volunteer Assistance Request	
Actors	User, Volunteer
Goal	Enable users to request trip assistance and connect with a volunteer.
Preconditions	User must be signed in, and available volunteers should exist.
Trigger	User selects the "Request Volunteer" option.
Main Flow	1) Enter trip details. 2) System searches for matching volunteers. 3) Request is sent to available volunteers. 4) First volunteer to accept is assigned.
Alternative Flows	A1: No volunteer accepts the request. A2: Assigned volunteer cancels. A3: GPS data unavailable.
Postconditions	Volunteer is assigned, and status is updated accordingly.
Business Rules	The first volunteer to accept will automatically be assigned.

Sanad Platform: Request Acceptance or Rejection	
Actors	Volunteer, Interpreter
Goal	Allow providers to accept or decline user requests.
Preconditions	Provider must be logged in and available.
Trigger	A new request notification is received.
Main Flow	1) Provider views request details. 2) Provider selects Accept or Reject. 3) System updates request status. 4) Notifications are sent to the user.
Alternative Flows	A1: Time conflict occurs. A2: Response timeout. A3: Provider account inactive.
Postconditions	Request status is updated to Accepted or Rejected.
Business Rules	Providers must respond within a defined time limit.

Sanad Platform: GPS Location Tracking	
Actors	User, Volunteer
Goal	Enable real-time GPS tracking during active service.
Preconditions	Request must be accepted, and location access granted.
Trigger	Tracking begins when the service starts.
Main Flow	1) Enable GPS tracking.2) Display live map.3) Update user and volunteer locations in real-time.4) End tracking upon service completion.
Alternative Flows	A1: GPS disabled.A2: Network connection lost.
Postconditions	Tracking session ends, and location data is securely saved.
Business Rules	GPS sharing is only permitted during the active service period.
System	Sanad Platform

3.3.4 <Content Access and Library>

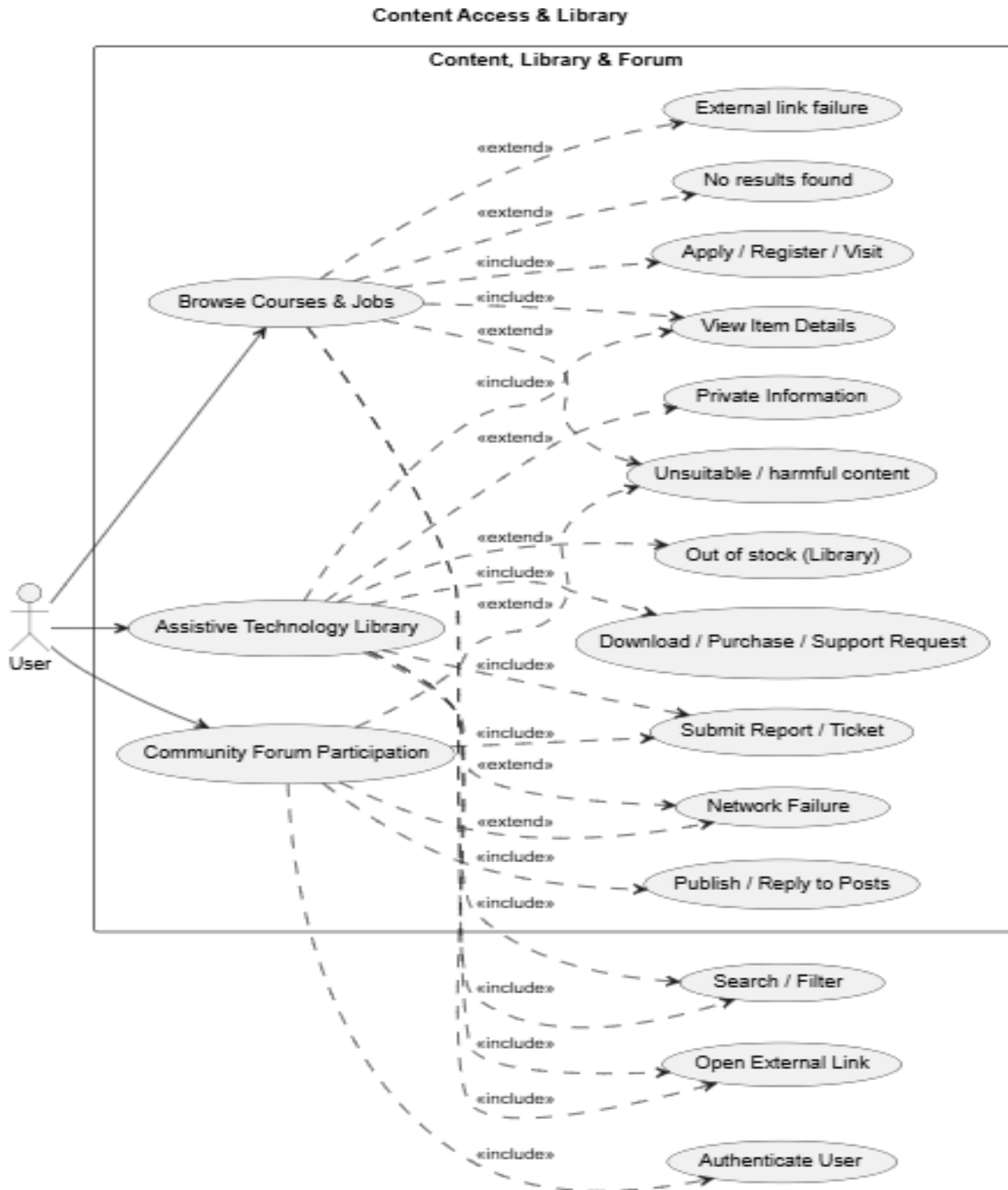


Figure 3.3.4 – Content Access and Library

Sanad Platform: Browse Courses & Jobs	
Actors	User
Goal	Give the user the option to open the relevant external links and find training programs and employment openings that are easily accessible.
Preconditions	• The user is logged in (per policy, guest browsing is permitted). • Course and job data sources are set up and accessible. • There is internet access.
Trigger	The user starts a search or filter or navigates to the Courses & Jobs page.
Main Flow	1) Choose the "Courses/Jobs" tab. 2) Configure the filters (date, skill, location/remote, and accessibility). 3) The system displays badged results. 4) Details of the open item. 5) Select the Apply/Register/Visit link. 6) (Optional) Send a notification or add to the calendar.
Alternative Flows / Exceptions	A1) No outcomes → Recommend calming filters. A2) When an external link fails, an error message and a copy-link option are displayed. A3) Unsuitable content → The report opens a ticket for moderation. A4) Wants to apply but is not logged in → Prompt Create an account or log in.
Postconditions	An optional bookmark or calendar entry is made when an item is viewed or an external link is opened.
Business Rules	Give verified and accessible content priority; mark content as "accessible" and automatically hide expired items.
Non-Functional Requirements (NFR)	Page load ≤ 2s; WCAG-compliant UI; privacy-respecting analytics.
Notes	Use safe redirects to open external links, and if necessary, display a privacy notice.

Sanad Platform: Assistive Technology Library	
Actors	User
Goal	Browse assistive technology apps and products, read reviews and specs, and find links for downloads, purchases, and support.
Preconditions	• Product details, media, and reviews are added to the catalog. • Configure endpoints for purchase, storage, and download.
Trigger	The user searches for a product or category or opens the library.
Main Flow	1) Select a category. 2) Products are listed by the system (price, platform, compatibility, tags). 3) Go to the product page. 4) Select Purchase/Download or Submit a Support Request. 5) Send a confirmation after redirecting to the store, downloading, or opening the support form.

Alternative Flows / Exceptions	A1) Out of stock → "Notify me" or suggest comparable options. A2) Provide alternate links in case the app store link is broken. A3) Personal fitting is required → route to the supplier or partner.
Postconditions	The user is informed when the external navigation is finished or a support ticket is created.
Business Rules	Only verified vendors are displayed; reviews are moderated; compatibility warnings are displayed.
Non-Functional Requirements (NFR)	Product page open time ≤ 2.5 seconds; alt text and bilingual labels; secure redirects.
Notes	Use safe redirects to open external links, and if necessary, display a privacy notice.

Sanad Platform: Community Forum Participation	
Actors	User
Goal	Use community moderation when posting new content or responding to already-existing threads.
Preconditions	• The user has been verified; the account is in good standing. • The terms and code of conduct are accepted. • Spam and abuse prevention measures are in place.
Trigger	The user clicks "New Post" or "Reply."
Main Flow	1) Write the content and title; include tags. 2) (Optional) Include links or images; use caution. 3) Press the "Publish" button. 4) The system shows posts or flags for examination. 5) Notify mentions and followers.
Alternative Flows / Exceptions	A1) Spam or toxic content found → Block publication; provide justification. A2) Private information → Make a change request. A3) Failure of the network → Save as Draft. A4) Edit/Delete → Permitted in a grace window following publication.
Postconditions	Posts are published, queued, or saved as drafts; notifications are sent or queued.
Business Rules	restrictions on length, media, and links; shadow moderation based on reports; and advanced privileges granted by reputation gates.
Non-Functional Requirements (NFR)	XSS/CSRF protection and URL sanitization; accessible editor; publish latency ≤ 1.5s.
Notes	Use safe redirects to open external links, and if necessary, display a privacy notice.

3.3.5 <Admin and System Monitoring>

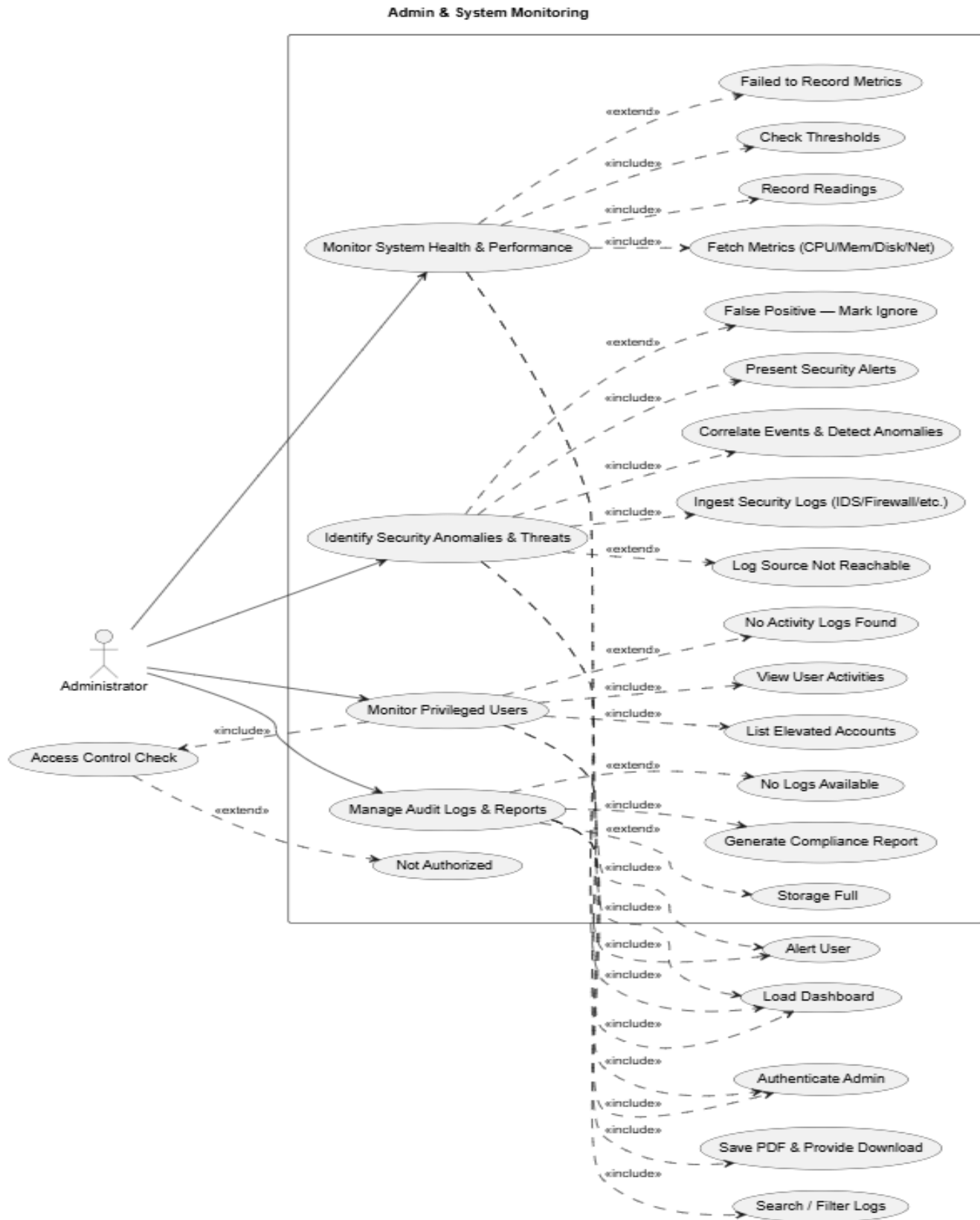


Figure3.3.5- Admin and System Monitoring

Sanad Platform :Monitoring Of Privileged Users	
Primary Actor	Administrator
Goal	Monitor the behavior of users with elevated privileges and identify any possible misuse.
Preconditions	The Administrator has logged into the system. The Administrator can read activity logs.
Postconditions	The activity has been logged in the system and is available for auditing.
Main Flow	1. The Administrator opens the dashboard. 2. The system shows a list of users with elevated privileges. 3. The Administrator selects a user to show activities. 4. The system shows logged logon operations, commands, orders, and events
Alternative Flow	A1: No activity logs found → The system displays an alert: "No activity has been logged." A2: Administrator does not have permission → The system displays a message: "Not authorized."

Sanad Platform :Monitor System Health & Performance	
Primary Actor	Administrator
Goal	To observe the state and performance of the system so advocates that failure is not imminent.
Preconditions	The system is running and collecting performance data.
Postconditions	The user is notified if it is in the red.
Main Flow	1. Admin opens the performance monitoring dashboard 2. The system displays the performance metrics: CPU / Memory / Disk / CPU. 3. The system records the current readings. 4. The system displays an alert if the thresholds are exceeded.
Alternative Flow	A1: System Fails to Record Data → The system records internal error "No Metrics Available".

Sanad Platform : Identify Security Anomalies & Threats	
Primary Actor	Administrator
Goal	Detect security threats or abnormal activities.
Preconditions	Logs / IDS / Firewall / Core integration with system is present.
Postconditions	The detected incidents are logged into the System Incident Database.
Main Flow	1. Admin opens Security Analytics. 2. The system receives logs from security sources. 3. The system uses correlation rules to detect anomalies. 4. The system presents the alert on the screen.
Alternative Flow	A1: False Positive → Admin selects "Ignore." A2: Failed Connection to Log Source → System alerts: "Source Not Reachable. "

Sanad Platform :Management of Audit Logs and Reports	
Primary Actor	Administrator
Goal	Administer logs of systems and generate compliance-based reports.
Preconditions	The administrator is logged in with permission to access logs.
Postconditions	A report has been raised and stored in the repository.
Main Flow	1. Admin selects Audit Logs. 2. The system shows the logs (filter / search). 3. Admin selects "Generate Compliance Report." 4. The system saves reports as PDF and provides a link to download."

Alternative Flow

A1: No Logs Available → The system says "No logs available";
A2: Storage Full → The system says "Storage full."

3.4 CLASSES / OBJECTS

These categories represent the main components of the Sanad system, where data is stored and processed.

3.4.1 <User>

The User class represents any individual who interacts with the Sanad Platform. Including individuals with disabilities, their families, volunteers and interpreters. This class stores personal information required for authentication and profile customization as shown in figure [3.4.1]. The User object also maintains optional accessibility-related information to support interface adaptation based on disability type.

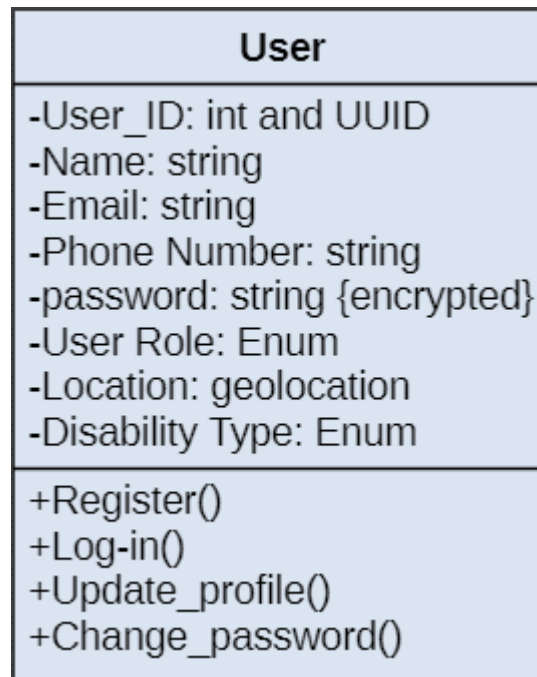


Figure 3.4.1 – User Class

Attributes			
Attribute	Description	Type	Constraints
User_ID	Unique identifier for the user.	Integer / UUID	Auto-generated and required.
Name	Full name of the user.	String	Required.
Email	User email for login and notifications.	String	Optional (required if phone is not provided).
Phone Number	User phone number for login and verification.	String	Optional (required if email is not provided).
Password	Encrypted password for authentication.	String	Required.
User Role	Defines role: • Individuals with disabilities. • Individuals with disability families. • Volunteers. • Interpreters.	Enum	Required.
Location	The user's geographical location (for the matching service).	Geolocation	Required.
Disability Type	Specifies disability category for accessibility customization.	Enum	Optional.

Operations (Methods)	
Method	Description
Register()	Creates a new account with provided information.
Log-in()	Authenticates the user using email or phone number and password.
Update_profile()	Allows modification of personal information.
Change_password()	Allows users to update password before verification.

3.4.2 <Location>

The **Location** class represents the geographical information of users and services within the Sanad Platform. It is primarily used for **service matching**, **navigation**, and **regional filtering** of available resources (such as volunteers, interpreters, or nearby assistive services). This class ensures that all stored location data follows a consistent and accurate format to support accessibility and real-time operations.

Location
-Location_ID : UUID -User_ID : FK(UuerID) -Latitude : Decimal -Longitude : Decima/ -Address : String -City : String -Region : String -Country : String -PostalCode : String
+setLocation() +getLocation() +findNearbyServices()

Figure 3.4.2 - Location class

Attribute	Description	Type	Constraints
Location_ID	A unique identifier for each location record.	Integer / UUID	Auto-generated and required.
User_ID	Identifier of the user associated with the location.	FK (UserID)	Must exist in the User table and required.
Latitude	The geographic latitude coordinate of the user or service.	Decimal	Required and must follow valid GPS format.
Longitude	The geographic longitude coordinate of the user or service.	Decimal	Required and must follow valid GPS format.
Address	The human-readable address or description of the location.	String	Optional but recommended for clarity.
City	The city or locality name.	String	Required.

Region	The region, province, or governorate name.	String	Optional.
Country	The country where the user or service is located.	String	Required.
PostalCode	Postal or ZIP code for additional precision.	String	Optional.

Method	Description
setLocation()	Records or updates the user's current location.
getLocation()	Retrieves the stored location details for a specific user.
findNearbyServices()	Returns a list of interpreters, volunteers, or events available near the user's coordinates.
validateCoordinates()	Ensures the entered latitude and longitude values are valid and within supported range.

3.4.3 <Review>

The Review class allows users to provide feedback, comments, and ratings on services or assistive technology items within the Sanad Platform. It supports maintaining transparency, improving service quality, and enhancing user trust by enabling community-based evaluation.

Review
-Review_ID : UUID -User_ID : FK(UserID) -Target_ID : UUID -TargetType : Enum -Rating : Integer -Comment : Text -Timestamp : DateTime -Status : Enum
+addReview() +editReview() +deleteReview() +getReviewsByTarget() +calculateAverageRating

Figure 3.4.3 – Review class

Attribute	Description	Type	Constraints
Review_ID	A unique identifier for each review.	Integer / UUID	Auto-generated and required.
User_ID	The ID of the user who submitted the review.	FK (UserID)	Must exist in the User table and required.
Target_ID	The identifier of the reviewed entity (e.g., ServiceBookingID, AssistiveTechnologyID).	UUID / Integer	Must correspond to a valid reviewed object.
TargetType	Specifies what is being reviewed (e.g., “Service”, “AssistiveTechnology”, “JobCourse”).	Enum	Required.
Rating	The numeric rating value given by the user (1–5).	Integer	Required and must be between 1 and 5.
Comment	The text feedback provided by the user.	Text	Optional but recommended.
Timestamp	The date and time when the review was created.	DateTime	Auto-generated and required.
Status	The moderation status of the review (Approved, Pending, Reported).	Enum	Default = Pending.

Method	Description
addReview()	Allows a verified user to submit a new review for a completed service or product.
editReview()	Enables the user to modify their existing review before approval.
deleteReview()	Removes a review from the system upon user request or admin moderation.
getReviewsByTarget()	Retrieves all reviews and ratings related to a specific target (service, course, or technology).
calculateAverageRating()	Computes the average rating score for a given target item based on all submitted reviews.

3.4.4 <ServiceBooking>

Attributes

Attribute	Description	Type
BookingID	A unique identifier for each booking (interpret as escort).	Integer / UUID
ClientID	User ID of the service requester.	FK (UserID)
ProviderID	Service provider ID (translator or volunteer)	FK (UserID)

ServiceType	Type of service (Interpreter / Volunteer).	Enum
StartTime	The date and time the service will start.	Timestamp
Duration	Expected service duration (in hours or minutes).	Integer
Status	Booking status (Pending, Confirmed, InProgress, Completed, Cancelled).	Enum
Video Call Link	The link to the interpreter's video call room (is created upon confirmed booking).	URL

Operations

Operation	Description
CreateBooking()	Create a new booking process based on user input.
ConfirmBooking()	Booking confirmation after matching with the service provider.
JoinVideoCall()	Join the video room for the interpreter session.
UpdateStatus()	Updating the booking status (e.g., moving from Pending to Completed).

3.4.5 <ForumPost >

The ForumPost class model user generated discussion content within the Sanad Platform. This ranges from new topics and replies to purely informative posts, all created by registered users. This class will store essential details regarding the post title, content, author, category, and publication timestamp, as shown in figure [3.4.5]. It will also keep moderation-related attributes to support content review, reporting, and community safety.

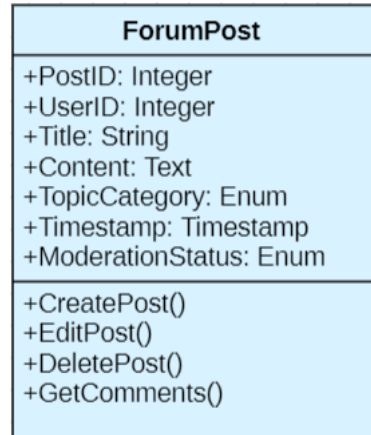


Figure 3.4.5 – ForumPost Class

Attribute	Description	Type	Constraint
PostID (Primary Key)	A unique identifier for the post or forum topic.	Integer/UUID	Must be unique
UserID (Foreign Key)	Identifier of the user who created the post.	FK (UserID)	Must exist in the User table (Valid FK)
Title	Title of the forum post.	String	Required.
Content	The text content of the post.	Text	Required.
TopicCategory	The category of the post (exp: Tips, Experiences, Advice, Technology).	Enum	-
Timestamp	The date and time of the post was published.	Timestamp	-

ModerationStatus	The moderation status of the post (exp: Approved, Reported).	Enum	-
Operations	Actions that can be performed on the post (CreatePost(), EditPost(), DeletePost(), GetComments()).		

Operations (Methods)	
Method	Description
CreatePost()	Creates a new forum post .
EditPost()	Allow post creator to edit their post .
DeletePost()	Remove post .
GetComments()	Retrieve comment from a post.

3.4.6 <JobCourse>

This class represents training courses and programs to prepare you for work, which you find in the Learning, Employment, and Technology component of the Sanad platform. It holds a training course's title, description, schedule, and delivery method (online/on-site). This class is responsible for enrolling users, and sending notifications for updates or upcoming training and readiness course.

JobCourse
-id: UUID -title: String -type: Enum -category: String -provider: String -location: String -accessibilityTags: List -startDate: Date -endDate: Date -isSaved: Boolean -externalLink: URL -status: Enum
+saveListing() +removeSavedListing() +apply() +share() +updatestatus()

Figure 3.4.6 - JobCourse

Attribute	Type	Description
id	UUID / String	Each listing item's unique identifier
title	String	The job posting or course name.
type	Enum (Course, Job)	specifies if the entry is for a job or a course.
category	String	The listing's classification (e.g., IT, Business, Accessibility Skills).
description	Text	A thorough explanation of the listing.
provider	String	Name of the employer or course provider.
location	String	Country, city, or "Remote"
accessibilityTags	List<String>	badges that support accessibility, like "Sign Language Available" and "Wheelchair Accessible."
startDate	Date	The opportunity or course's start date.
endDate	Date	Application/enrollment closing date.
isSaved	Boolean	indicates whether the user has bookmarked or saved the listing.
externalLink	URL	Use this link to register or apply.
status	Enum (Active, Expired, Closed)	Opportunity status.

Operation	Description
saveListing()	Add this listing to the user's list of saved items.
removeSavedListing()	Take the listing out of the saved items.
apply()	takes the user to an external application or registration page.
share()	creates a direct link or a shareable link for social media.
updateStatus()	automatically modifies the status according to the start and end dates.

3.4.7 <AssistiveTechnology>

This class signifies assistive tools and technologies provided in the platform, including screen reader and accessibility settings, as well as compatible devices. It will maintain data about each assistive function and will activate each instrument based on user preference for accessibility to benefit and foster an inclusive user experience.

AssistiveTechnology
-id: UUID -name: String -type: Enum -targetDisability: String -description: Text -platformCompatibility: List -price: int -mediaImages: List<URL> -rating: Float -externalLink: URL -isFeatured: Boolean -status: Enum
+viewDetails() +addReview() +markAsFavorite() +removeFavorite() +openDownloadLink() +updateStatus()

Figure 3.4.7 Assistive Technology

Attribute	Type	Description
id	UUID / String	Every assistive technology item has a unique identification number.
name	String	Name of the technology or product.
type	Enum (Device, Software, MobileApp)	explains the technology's category.
targetDisability	String	type of supported disability (e.g., cognitive, mobility, visual, hearing).
description	Text	A thorough explanation of the application or product.
platformCompatibility	List<String>	platforms that are supported (Android, iOS, Windows, etc.).
price	Decimal / String	may be "Free," "One-time purchase," or "Subscription."
mediaImages	List<URL>	URLs for demo images or screenshots.
rating	Float	Average user rating (1.0–5.0).
externalLink	URL	redirect the download or

		purchase link.
isFeatured	Boolean	identifies the item as recommended or highlighted.
status	Enum (Active, Unavailable, Deprecated)	Status of availability.

Operation	Description
viewDetails()	returns all of the technology's details, including reviews, pictures, and descriptions.
addReview()	enables verified users to leave reviews or ratings for this assistive technology product.
markAsFavorite()	adds the item to the user's list of favorites.
removeFavorite()	deletes the item from the list of favorites.
openDownloadLink()	takes users to the download link or official store.
updateStatus()	updates the availability status according to the life cycle of the product.

3.4.8 <Notification>

The Notification class is responsible for sending alerts, reminders, and confirmation messages to users within the Sanad Platform. Notifications are used to inform users about account-related actions (such as password change verification), new messages, upcoming appointments or other system events. Notifications may be delivered through in-app alerts, email or SMS depending on the user's registered contact information. Figure [3.4.8].

Notification
-Notification_ID: int and UUID -Message_content: string -Notification_type: string -Delivery_method: string -Timestamp: datetime -Status: string
+Send_Notification() +Update_Status() +Get_Notification_History()

Figure 3.4.8 – Notification Class

Attributes			
Attribute	Description	Type	Constraints
Notification_ID	Unique identifier for each notification.	Integer	Auto-generated and required.
User_ID	Identifies the user receiving the notification.	Integer	Must exist in User table (foreign key) and required.
Message_content	The body text of the notification.	String	Cannot be empty and required.
Notification_type	The category of notification: • Verification. • Reminder. • System Alert.	String	Must match predefined notification types and required.
Delivery_method	How the notification is sent: • Email. • SMS. • In-App.	String	Must match allowed system-supported delivery methods and required.
Timestamp	The date and time when the notification was created or sent.	DateTime	Must follow system time format and required.
Status	Indicates the state: • Sent. • Delivered. • Read.	String	Must be one of: • Sent. • Delivered. • Read. Also required.

Operations (Methods)	
Method	Description
Send_Notification()	Sends the notification using the selected delivery method.
Update_Status()	Updates the notification status based on user interaction.
Get_Notification_History()	Retrieves all notifications for a given user.

3.4.9 <Admin Report>

The Admin Report class models the analytical and operational reports created by administrators in the Sanad Platform; such reports enable monitoring system activity, user engagement, quality of content, and performance of services. As seen in figure [3.4.9], the class maintains information like report type, date range, data output, and

identity of the admin generating it. Admin Report objects support oversight of the platform, decision-making, and long-term improvement of the system.

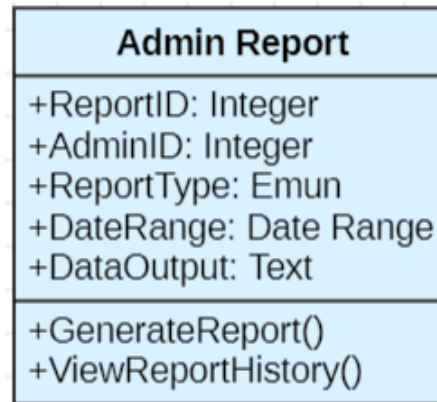


Figure 3.4.9 – Admin Report Class

Attribute	Description	Type	Constraint
ReportID (Primary Key)	A unique identifier for the report.	Integer / UUID	Must be unique
AdminID (Foreign Key)	Identifier of the admin who generated the report.	FK (UserID)	Must exist in the User table (Valid FK) and must belong to a user with Admin
ReportType	The type of report (exp: Financial, Performance, User Activity, Content Review).	Emun	-
DateRange	The date range is covered by the report.	Date Range	-
DataOutput	The generated report data (may be text or a downloadable file such as CSV/JSON).	Text / URL	-
Operation	Actions that can be performed on the report (exp: GenerateReport(), ViewReportHistory()).		-

Operations (Methods)	
Method	Description
GenerateReport()	Generate a report from the selected reportType and dateRange
ViewReportHistory()	Retrieves reports and or view names of reports from previously generated ones.

3.5 NON-FUNCTIONAL REQUIREMENTS

3.5.1 Performance

- **NFR-P1. Response Time** Under typical network conditions, the system should load all of the main screens (Home, Map, and Learning/Job listings) in less than three seconds (4G / Wi-Fi).
- **NFR-P2. API Response Time** Client requests must be answered by the backend API in less than 800 milliseconds (p95).
- **NFR-P3. Map Performance** After the Map module is opened, the accessibility layer loading and map rendering must be finished in less than one and a half seconds.
- **NFR-P4. Scalability** At least 10,000 users must be able to access the system at once without experiencing any performance issues.
- **NFR-P5. Database Performance** Read/write operations in databases must be completed in less than 500 milliseconds.
- **NFR-P6. Background Processing** Notification dispatch and listing updates are examples of scheduled tasks that must operate asynchronously without interfering with user interface functionality
- **NFR-P7. Pagination / Lazy Loading** No more than 25 listings will be retrieved per fetch by the system, which will use pagination or infinite scrolling.
- **NFR-P8. Caching** To cut down on repeated server requests, frequently accessed data (such as map markers and saved listings) should be cached.
- **NFR-P9. Upload Efficiency** The platform must allow for upload streaming and progress indicators when media content is uploaded.
- **NFR-P10. Error Handling** It is necessary to switch to a fallback lightweight view during periods of high load rather than permitting performance failure

3.5.2 Reliability

This requirement pertains to the system's ability of uninterrupted service or unauthorized/accidental compromise of data.

- 4 **Transaction success (NFR-R1):** The system must guarantee successful completion of nearly all functions, sustaining a failure rate of less than 0.01% for transactions to achieve a high level of data integrity and consistent results for all functions.
- **Recovery time (MTTR) (NFR-R2):** After any unforeseen failure or crash of the system, the platform must have full service restored within a maximum of 10 minutes to minimize downtime for users and meet important business continuity benchmarks.
- 5 **Call quality (NFR-R3):** For long-duration video communication, the dedicated video translation service will need to limit instability in communication sessions with the requirement that the dropped call ratio is less than 1% for all sessions that last longer than 5 minutes.

3.5.3 Availability

- **NFR-A1. System Uptime** With the exception of planned maintenance, the system must be accessible 99.5% of the time each year.
- **NFR-A2. Maintenance Window** Notifications of scheduled maintenance must be sent out at least 24 hours in advance.
- **NFR-A3. Redundancy / Backup** To prevent data loss in the event of a server failure, the system must maintain real-time data replication.
- **NFR-A4. Auto Recovery** In the event of a crash, critical services will automatically restart.
- **NFR-A5. Failover** The platform will switch to a backup instance in less than 30 seconds in the event of a failure.
- **NFR-A6. Offline Handling (Limited Mode)** When offline, the system will permit read-only access to cached content.
- **NFR-A7. Monitoring & Alerts** When there is downtime or a decline in performance, system monitoring will alert the support staff.
- **NFR-A8. Data Backup Frequency** incremental backups every four hours and daily full backups.
- **NFR-A9. Disaster Recovery (DRP)** After a significant incident, the system must return to normal operation within 24 hours.

3.5.4 Security

The system shall ensure comprehensive protection of user data, account information and communication channels through the implementation of secure access controls, standardized encryption mechanisms and reliable data handling practices. These measures are intended to maintain data confidentiality, integrity and availability across all platform components.

- **NFR-S1. Sensitive information**, including but not limited to user passwords and geographic location data, shall be encrypted during both storage (at rest) and transmission (in transit). Data transmission shall be conducted exclusively via the HTTPS protocol to prevent unauthorized interception. Compliance with this

requirement shall be verified through periodic security assessments and formal source code audit procedures.

- **NFR-S2. Furthermore**, the system should ensure the privacy and confidentiality of video-based interpretation services by enforcing end-to-end encryption, thereby preventing any third party, including system administrators, from accessing or monitoring transmitted audio or video content. Validation of this requirement shall be supported by encrypted traffic analysis and protocol verification reports.
- **NFR-S3.** to enhance account security, the system shall implement Two-Factor Authentication (2FA) for administrators, translators and volunteers. This additional security layer serves to mitigate the risks associated with credential theft or misuse. The effectiveness of this measure shall be monitored through the tracking of 2FA enrollment and usage rates among the designated user roles.
- **NFR-S4.** Additionally, the system shall employ robust protection against common security vulnerabilities, including Injection attacks and Cross-Site Scripting (XSS). This shall be achieved through rigorous input validation, data sanitization procedures and secure coding practices. System resilience shall be evaluated through scheduled vulnerability scans and penetration testing activities.

3.5.5 Maintainability

This section describes the ease with which the system can be modified, repaired and improved.

NFR.M1.Understandability

Description:

Code readability and proper documentation are key. It is expected that developers should apply consistent naming conventions and documentation standards.

Measurable Metric: Time to Understand Module: This is the average time it takes for a developer to understand a new or existing module.

NFR.M2. Repairability

Description:

It should provide the possibility of easy correction of minor defects (bugs), in order to promptly fix the problems.

Measurable Metric: Mean Time To Repair (MTTR): Minor bugs must be fixed within 48 hours of reporting.

NFR.M3 .Modularity

Description:

The system shall be designed on modular principles. A variation in one module should not require modification of another module unless they are directly related, such as dependent components.

Measurable Metric: Coupling Index: This describes the measure of interdependence between modules.

3.5.6 Portability

Describes the operational toughness of the system, in various environments.

- **Platform Compatibility (NFR-01):** The mobile application should work in its entirety on the two most recent major versions of the iOS and Android operating systems. Compatibility Testing Reports will evidence this implementation.
- **Cloud Readiness (NFR-02):** The backend architecture must use current purposely building containerization (Docker) to streamline deployment and migration to a major cloud service provider (AWS, Google Cloud, etc.). This is evidenced by the number of steps needed for cloud migration.

3.5.7 Inverse Requirements

This section specifies the behaviors and actions that the system must not allow. These requirements ensure system security, protect user data and prevent improper or unauthorized use of the platform.

Requirement	Priority
IR.1 The system must not allow any user to access system features without logging in.	High
IR.2 The system must not store any financial payment information (such as credit card numbers). All payment data must be securely transmitted directly to the external payment gateway without being saved in the system.	High

IR.3 The system must not store passwords in plain text; they must always be encrypted.	High
IR.4 The system must not crash when incorrect or unexpected input is entered; it should display clear error messages.	Medium
IR.5 The system must not send verification codes or notifications to invalid or unverified contact information.	Medium

3.5.8 Design Constraints

Identifier	Full description	Demands
Accessibility Standards (DC-1)	Compliance with WCAG 2.1 Level AA standards must be taken into account throughout the entire design and development process.	Make sure that all UI elements, content, and interactions adhere to WCAG 2.1 Level AA accessibility standards throughout both the design and development process.
Development Technology (DC-2)	The application must be built on a single platform (Flutter or React Native) for mobile applications.	To ensure a single codebase for development on both the Android and iOS platforms, you should use either Flutter or React Native as the development platform.
Language (DC-3)	The design and content will be in Arabic.	All user interface text, content, and design elements will be in Arabic as the primary language.

3.5.9 Safety Requirements

The platform works with individuals experiencing a disability and involves personally identifiable information, routing an individual, scheduling accessible service appointments, and volunteering/service, any sensitive information requires safety protocols.

The system must enforce the following safety protocols:

- **Protection of Sensitive Activities**

The system shall require explicit user acknowledgement before performing any irreversible or sensitive activities, such as deleting on an account, cancelling scheduled and approved appointments, or submitting a new access location to the accessibility map.

- **Safe Handling of Video and Service Sessions**

Video call sessions between users and sign-language interpreters shall not be recorded, saved, or shared by the system unless both the user and signer consent.

- **Temporary Live-Location Sharing**

Live-location sharing can only occur during a service (interpreter or volunteer escort) session. Live-location access shall cease automatically when the service ends.

- **Volunteer/Interpreter Safety Controls**

Only verified volunteers and interpreters are allowed to accept service or escort requests. The system shall maintain logs of actions performed by service providers.

- **Data Backup and Recovery**

The system will maintain securely and periodically backed up data to avoid data loss and to enable continued service to customers if the system fails or unexpectedly powers down.

- **Content Safety and Safety of Community**

Users must also be able to report posts, inappropriate content or other concerning activity. Moderators will investigate the concerns raised and take corrective action.

- **Fail-Safe for Handling Error**

If the connection drops or there is an issue with an API connection to a service (maps or video call service functionality) then the system will prompt an alert to inform the user of the issue with fallback options such as retry and reschedule options.

- **Physical Safety Considerations**

When the user requests a volunteer escort service, the user must consent to that person's visibility of their profile and status to conduct awareness of personal safety prior to meeting.

The users' safety requirements ensure that the platform protects the user from the misuse of data, safe service exchange and communication and trust of users and service provider.

3.5.10 Business Rules

The following business rules govern the process of information, service and user interactions in the Sanad Platform:

- **Rule of Designating User Status**

Every user must state their status as part of their registration as to whether they are a Client, Volunteer, Interpreter and Administrator.

The system provides access and other permissions based on the user status.

- **Rule for Confirming Service Providers.**

Volunteers and Interpreters must verify their identity prior to being able to receive or accept service requests. Users who have not verified their identity may not provide assistance or accept appointments.

- **Rule for Content Moderation.**

Content that is viewable by the public, or would impact other users (i.e. updating, map location, accessibility ratings, and inspiring stories) must be reviewed and approved by the moderator prior to publishing.

- **Rule for Reporting and Community Safety**

Users may report abusive, damaging, offensive or misleading posts Moderation must take action (review, remove and/or request proof) within the defined time.

- **Rule for ratings and reviews.**

Only verified users may review/ or rate places or services, and each user may only contribute one review of place or service.

- **Appointment Activation Rule**

Once a service has been booked, (whether the service is for a volunteer escort or interpreter), it is active only when the appointment is accepted. The appointment status (Pending, Accepted, Completed or Canceled) should reflect the current appointment status.

- **Cancellation Rule**

Users can cancel their appointments 24 hours before the appointment starts. Cancellations made after this time will require administrator approval.

- **Critical Action Confirmation Rule**

Any action that will permanently delete data or post data that can be publicly viewed and can be shared with users (deleting an account, publishing the appointment location, cancelling an appointment that has been accepted, etc.) will require the user have to confirm that they still want to take the action.

- **Live Location Sharing Rule**

Users can only share their live location with the assigned service provider during the appointment, and access to the live location will expire once the service provider ends the service or the appointment is finished.

4. CHANGE MANAGEMENT PROCESS

Any changes, regardless of whether they were initiated by the client or the development team, must be vetted carefully against the strategic objectives and operational needs of the organization. The organizational context, timeline, budget, and quality realized by the project benefiting from change must also be subjected to quality assessment concerning the change, in order to ensure a comprehensive understanding of the impact. Further, scope creep must be avoided, and project stability maintained, which requires a strict adherence to boundaries from a change management stance. All change requests must be considered against the relevant criteria for submission, review, authorization, implementation, and review. Well-defined control, reliable and active management, does minimize risk, protect project integrity, and facilitate project success.

4.1 Purpose

The intent of this process is to secure that all changes proposed are documented in the Service Request System (SRS). Changes will go through the full chain of approvals and formally documented in SRS before implementation can start. Having a systematic and documented mechanism allows for the integrity, consistency, and sustainability of the project; by having a clearly documented history of change, it allows for tracking and accountability. Following these standards allows the organization to ensure that there are no ad hoc or undocumented changes, reduced risk, and confirm that all change aligns to the objectives of the project and the organization.

4.2 Procedures

Step	Responsible	Action
Change Request (CR)	Clinic/Developer	Complete a change request form, detailing the rationale for the proposed change and its effect.
Impact analysis	Project manager & Technical teams	Consider the effect of the change on the schedule, costs, and any other aspects of the project.
Approval/Rejection	Change control board (CCB)	Consider the analysis and decide to approve or reject the proposed change.
Documentation	Project coordinator	Update the SRS to include modification history to track approved changes.
Implementation	Development team	Take actions to implement the approved change per the process with all its complete details.
Quality Assurance (QA)	QA team	Confirm the change satisfies the requirement and make adjustments if necessary prior to final release.

4.3 Tools

- **Version Control**

1. Git: In your comments, you noted that Git is a widely used version control system for code change management, branching, and rollback.
2. GitHub/GitLab/Bitbucket: Platforms that host Git repositories, enable code reviews, and support collaboration.

- **Change Tracking and Issue Management**

1. Jira: A widely used tool for tracking change requests, bugs, and project tasks.
2. Trello: A visual task management tool that provides tracking requests with boards, lists, and cards.
3. Asana: A task/project management tool that can be used for tracking change request and workflows.

- **Other Tools/Practices**

1. Configuration Management Tools: Tools such as Ansible, Puppet, or Chef to manage the infrastructure and deployment configurations.
2. Documentation Tools: Confluence, SharePoint, or Google Docs for comprehensive documentation.
3. Automated Testing and CI/CD: Jenkins, Travis CI, or GitLab CI to support continuous and automated testing and delivery.

- **Change Management Frameworks**

1. Frameworks such as ITIL or COBIT to formalize change management processes within an organization.